

习题 10.1

张志聪

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方法 1: 四个图形的面积之和, 证明略。

方法 2 (三角函数推荐): 利用变量替换, 函数的奇偶性。

按封闭曲线计算公式

$$\begin{aligned} A &= \left| \int_{\alpha}^{\beta} y(t)x'(t)dt \right| \\ &= \left| \int_0^{2\pi} a\sin^3 t(-3a\cos^2 t\sin t dt) \right| \\ &= \left| -3a^2 \int_0^{2\pi} \sin^4 t \cos^2 t dt \right| \\ &= 3a^2 \left| \int_0^{2\pi} \sin^4 t \cos^2 t dt \right| \end{aligned}$$

令 $u = t - \pi, t: 0 \rightarrow 2\pi, u: -\pi \rightarrow \pi, dt = du$, 于是

$$\begin{aligned} \int_0^{2\pi} \sin^4 t \cos^2 t dt &= \int_{-\pi}^{\pi} \sin^4(u + \pi) \cos^2(u + \pi) du \\ &= \int_{-\pi}^{\pi} \sin^4(u + \pi) \cos^2(u + \pi) du \\ &= \int_{-\pi}^{\pi} \sin^4 u \cos^2 u du \end{aligned}$$

因为 $\sin u$ 是奇函数, 所以 $\sin^4 u$ 是偶函数, $\cos u$ 是偶函数, 所以 $\cos^2 u$ 是偶函数, 于是 $\sin^4 u \cos^2 u$ 是偶函数。我们有

$$\int_{-\pi}^{\pi} \sin^4 u \cos^2 u du = 2 \int_0^{\pi} \sin^4 u \cos^2 u du$$

令 $z = u - \frac{\pi}{2}, u: 0 \rightarrow \pi, z: -\frac{\pi}{2} \rightarrow \frac{\pi}{2}, du = dz,$

$$\begin{aligned}\int_0^\pi \sin^4 u \cos^2 u du &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^4(z + \frac{\pi}{2}) \cos^2(z + \frac{\pi}{2}) dz \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^4 z \sin^2 z dz\end{aligned}$$

类似地, $\cos^4 z \sin^2 z$ 是偶函数, 我们有

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^4 z \sin^2 z dz = 2 \int_0^{\frac{\pi}{2}} \cos^4 z \sin^2 z dz$$

综上,

$$\begin{aligned}3a^2 \left| \int_0^{2\pi} \sin^4 t \cos^2 t dt \right| &= 3a^2 \left| 4 \int_0^{\frac{\pi}{2}} \cos^4 z \sin^2 z dz \right| \\ &= 12a^2 \left| \int_0^{\frac{\pi}{2}} \cos^4 z \sin^2 z dz \right|\end{aligned}$$

由第九章 §5 习题 8 可知

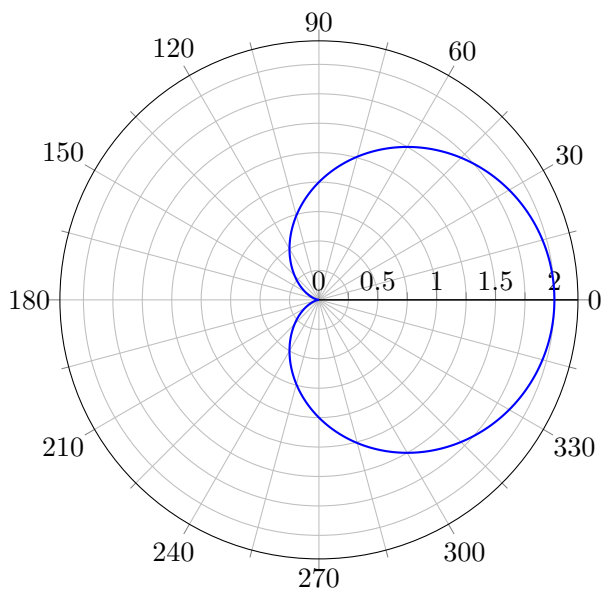
$$\begin{aligned}J(4, 2) &= \frac{1}{2} J(2, 2) \\ &= \frac{1}{2} \frac{1}{4} J(0, 2) \\ &= \frac{1}{2} \frac{1}{4} \frac{1}{2} J(0, 0) \\ &= \frac{1}{16} \frac{\pi}{2} \\ &= \frac{\pi}{32}\end{aligned}$$

所以

$$\begin{aligned}12a^2 \left| \int_0^{\frac{\pi}{2}} \cos^4 z \sin^2 z dz \right| &= 12a^2 \frac{\pi}{32} \\ &= \frac{3}{8} a^2 \pi\end{aligned}$$

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$r = 1(1 + \cos(x))$, 以 $a = 1$ 为例

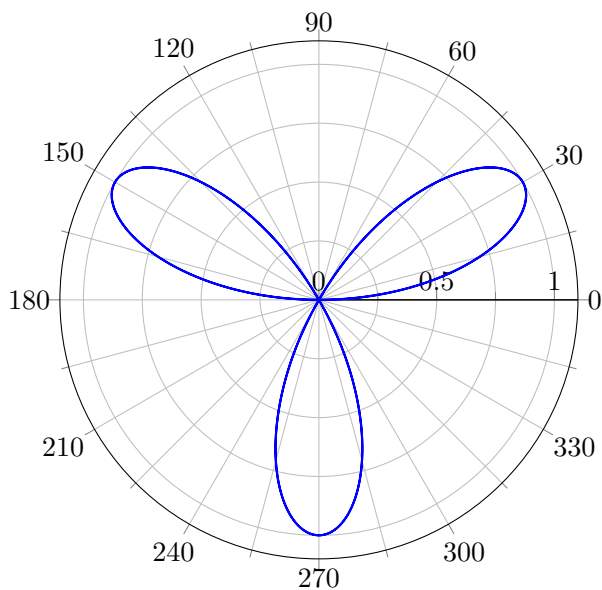


极坐标的面积:

$$\begin{aligned}
 A &= \frac{1}{2} \int_0^{2\pi} r^2(\theta) d\theta \\
 &= \frac{1}{2} \int_0^{2\pi} a^2 (1 + \cos\theta)^2 d\theta \\
 &= \frac{a^2}{2} \int_0^{2\pi} 1 + \cos^2\theta + 2\cos\theta d\theta \\
 &= \frac{a^2}{2} \left[\int_0^{2\pi} 1 d\theta + \int_0^{2\pi} \cos^2\theta d\theta + 2 \int_0^{2\pi} \cos\theta d\theta \right] \\
 &= \frac{a^2}{2} \left[2\pi + \int_0^{2\pi} \frac{1}{2} (1 + \cos 2\theta) d\theta + 2 \sin\theta \Big|_0^{2\pi} \right] \\
 &= \frac{a^2}{2} \left[2\pi + \frac{1}{2} \left(\theta + \frac{1}{2} \sin 2\theta \right) \Big|_0^{2\pi} + 0 \right] \\
 &= \frac{a^2}{2} [2\pi + \pi + 0] \\
 &= \frac{3}{2} a^2 \pi
 \end{aligned}$$

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$r = 1 \sin(3\theta)$, 以 $a = 1$ 为例



通过画图过程可知积分上下限为 $0 \rightarrow \pi$ (注: 直接看图是看不出来的)。

$$\begin{aligned}
 A &= \frac{1}{2} \int_0^{\pi} r^2(\theta) d\theta \\
 &= \frac{1}{2} \int_0^{\pi} r^2(\theta) d\theta \\
 &= \frac{1}{2} \int_0^{\pi} a^2 \sin^2 3\theta d\theta \\
 &= \frac{a^2}{2} \int_0^{\pi} \frac{1}{2} (1 - \cos 6\theta) d\theta \\
 &= \frac{a^2}{4} \int_0^{\pi} 1 - \cos 6\theta d\theta \\
 &= \frac{a^2}{4} \left(x - \frac{1}{6} \sin 6\theta \right) \Big|_0^{\pi} \\
 &= \frac{a^2}{4} \pi
 \end{aligned}$$

注: 玫瑰线

$$r = a \sin n\theta (a > 0)$$

当 n 为奇数, 闭合周期 π ;
 当 n 为偶数, 闭合周期 2π .

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显然 $x: 0 \rightarrow a$, 否则等式不成立。

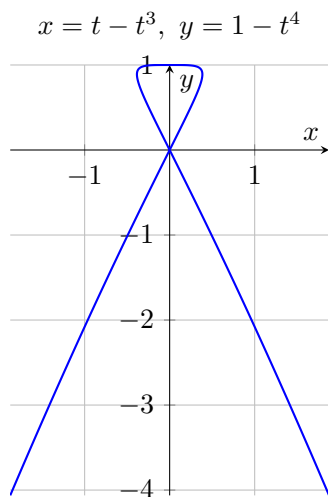
计算 y 关于 x 的表达式:

$$\begin{aligned}\sqrt{\frac{x}{a}} + \sqrt{\frac{y}{b}} &= 1 \\ \sqrt{\frac{y}{b}} &= 1 - \sqrt{\frac{x}{a}} \\ \frac{y}{b} &= 1 + \frac{x}{a} - 2\sqrt{\frac{x}{a}} \\ y &= b + \frac{b}{a}x - 2b\sqrt{\frac{x}{a}}\end{aligned}$$

于是, 面积为

$$\begin{aligned}A &= \int_0^a b + \frac{b}{a}x - 2b\sqrt{\frac{x}{a}} dx \\ &= \left(bx + \frac{b}{2a}x^2 - \frac{2b}{\sqrt{a}} \frac{2}{3}x^{\frac{3}{2}} \right) \Big|_0^a \\ &= \left(bx + \frac{b}{2a}x^2 - \frac{4b}{3\sqrt{a}}x^{\frac{3}{2}} \right) \Big|_0^a \\ &= ab + \frac{1}{2}ab - \frac{4}{3}ab - 0 \\ &= \frac{1}{6}ab\end{aligned}$$

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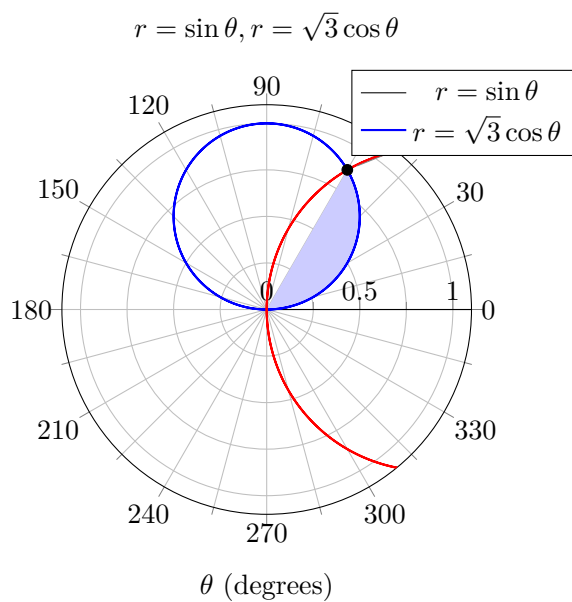


为什么这么画:<https://www.bilibili.com/video/BV1jW4y1S7z5?t=476.0&p=153>

因为 $x(-1) = x(1)$, 于是

$$\begin{aligned} A &= \left| \int_{-1}^1 y(t) x'(t) dt \right| \\ &= \left| \int_{-1}^1 (1 - t^4)(1 - 3t^2) dt \right| \\ &= \frac{16}{35} \end{aligned}$$

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对工具绘图用的不太好，右边的圆没有显示完整。
两个极坐标曲线的交点为

$$\sin \theta = \sqrt{3} \cos \theta$$

$$\tan \theta = \sqrt{3}$$

$$\theta = \frac{\pi}{3}$$

公共部分是两个部分的和

$$\begin{aligned} A &= \frac{1}{2} \int_0^{\frac{\pi}{3}} (\sin\theta)^2 d\theta + \frac{1}{2} \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} (\sqrt{3}\cos\theta)^2 d\theta \\ &= \frac{1}{2} \int_0^{\frac{\pi}{3}} \sin^2\theta d\theta + \frac{1}{2} \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} 3\cos^2\theta d\theta \\ &= \frac{1}{4} \int_0^{\frac{\pi}{3}} (1 - \cos 2\theta) d\theta + \frac{3}{4} \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} (1 + \cos 2\theta) d\theta \\ &= \frac{1}{4} \left(\theta - \frac{1}{2} \sin 2\theta \right) \Big|_0^{\frac{\pi}{3}} + \frac{3}{4} \left(\theta + \frac{1}{2} \sin 2\theta \right) \Big|_{\frac{\pi}{3}}^{\frac{\pi}{2}} \\ &= \frac{1}{4} \left(\frac{\pi}{3} - \frac{\sqrt{3}}{4} - 0 \right) + \frac{3}{4} \left(\frac{\pi}{6} - \frac{\sqrt{3}}{4} \right) \\ &= \frac{5\pi}{24} - \frac{\sqrt{3}}{4} \end{aligned}$$